

Mingyu Park

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Research Interests

- Design and controlled synthesis of porous materials for heterogeneous catalysis
- Engineering of active sites and transport environments for gas conversion and separations

Education

Seoul National University (SNU) *Mar. 2020 – Feb. 2026*
B.S. Candidate, Chemical and Biological Engineering *(Expected)*

- **GPA: 4.20/4.30 (3.96/4.00)** **Major: 4.26/4.30 (4.00/4.00)**

University of California, Berkeley *Jan. 2025 – May 2025*
UCEAP Study Abroad Program, Chemical Engineering

- **Major GPA: 4.00/4.00** (16 units; A+ in all four upper-division ChemE/Chem courses)

Seoul Science High School *Mar. 2017 – Feb. 2020*
High School for Gifted Students in Science and Mathematics

- **GPA: 4.28/4.30**; Admitted with highest distinction. (Rank 1/123)

Research Experience

Heterogeneous Catalysts Design Lab, SNU *Sep. 2025 – Present*
Undergraduate Researcher, PI: Prof. Jong Hun Kang

- Systematically tuning graphene oxide synthesis parameters to build predictive links between synthesis conditions and material properties. Investigating how specific surface oxygen groups govern gas-evolution behavior during thermal reduction using DSC, XPS, TPR-MS and TGA.

Katz Group, University of California, Berkeley *Feb. 2025 – Aug. 2025*
Undergraduate Researcher, PI: Prof. Alexander Katz

- Project 1: Synthesized TMSCl-silanized Na-FAU zeolites and showed that the surface modification preserves the framework integrity and H₂O/CO₂ adsorption capacity (via XRD, BET, TGA, and CHNS) while selectively suppressing H₂O adsorption kinetics under humid conditions, as confirmed by *in situ* DRIFTS.
- Project 2: Adapted the silanization strategy to HZSM-5 zeolites, which effectively inhibited external phosphorus deposition, suggesting a potential route toward selective interior phosphorus placement without the use of expensive OSDAs (organic structure-directing agents).

Machine Intelligence & Comp. Chem. Lab, SNU *Mar. 2024 – Jan. 2025*
Student-Directed Research Program, PI: Prof. Yousung Jung

- Developed a DFT-informed ML framework (Python, Q-Chem) to predict stereoselectivity in Grubbs olefin metathesis, identifying key molecular descriptors governing selectivity.
- Awarded the **Grand Prize in the Student Independent Research Awards** at SNU.

Publications & Presentations

Mingyu Park[†], Hwangho Lee[†], John Tengco, Alexander Katz^{*}. Monolayer TMSCl Silanization of FAU Zeolites Enables Kinetic Enhancement of CO₂ Adsorption Under Humid Conditions

- 2 Poster presentations: CeRCaS 2025 Spring Meeting (May 2025); UC Berkeley Undergraduate Research Summer Symposium (Aug. 2025).
- Manuscript in preparation for submission.

Honors & Awards

Presidential Science Scholarship, Korea Student Aid Foundation *Mar. 2020 – Present*
4-year full tuition and stipends (\$5,500 × 8); 93 undergraduates selected nationwide.

KorUS Educational Exchange Initiative for Youth in STEM, MOTIE *Nov. 2024*
One-time \$9,000 stipend; 102 exchange students selected nationwide.

Minister of Trade, Industry and Energy Award, MOTIE *Oct. 2025*
Awarded for the most outstanding achievement among Korea-U.S. STEM exchange students.

KICHe President's Award, Environmental Safety Design Competition *Oct. 2025*
Awarded 2nd place for designing a benzene/hydrogen production process from PET pyrolysis.

Grand Prize, Student Independent Research Awards, SNU *Feb. 2025*

Honorable Mention, Samsung Humantech Paper Award *Feb. 2020*
Awarded \$1,000 for a paper on synthesizing sustainable epoxy adhesives from waste lard oil.

National Team Candidate, Chemistry Olympiad, Korean Chemical Society *2019*

3x Gold Medalist, Korea Undergraduate Math. Competition *2020, 2021, 2023*

Teaching & Professional Experience

Air Intelligence Operator, Republic of Korea Air Force *Apr. 2022 – Jan. 2024*
- Managed intelligence data; Also served as the squadron's Senior Soldier Representative, mediating communication between officers and enlisted personnel and supporting unit decisions.

Volunteer Tutor, Community Learning Program *2021 – 2022, 2024 – Present*
- Dedicated 200+ hours to tutoring Chemistry and Mathematics for underserved students, providing academic support to bridge educational gaps.

Academic Tutor, SNU *Sep. 2024 – Present*
- Provided weekly academic support in Korean and English for undergraduate courses: General Chemistry (Fall 2024, Fall 2025) and Separation Processes (Fall 2025).

Skills

Synthesis & Reaction Engineering: Flow Reactor Setup, Hydrothermal & Organic Synthesis
Characterization: *in situ* DRIFTS, XRD, BET, NMR, XPS, TGA, UV-Vis

Modeling & Simulation: Aspen Plus, AVEVA Pro/II, Q-Chem

Programming: Python, MATLAB, C++ **Languages:** Korean (Native), English (Fluent)

Activities & Societies

STEM: SNU Tomorrow's Engineers Membership *Apr. 2024 – Present*
Honor society of SNU College of Engineering, with 25 members selected annually
- Director of External Affairs Department: Organized networking events with other academic societies at SNU and coordinated the 15th homecoming event with alumni.

- Hosted 2 academic seminars for members, including a talk on introducing porous materials.

Dalisha: SNU Running Club *Sep. 2024 – Present*
- Pacemaker: Led weekly training and races to assist members in maintaining target paces.
- Finished 2025 Chuncheon Marathon (3:41:59) and MBN Half Marathon (1:37:24).